

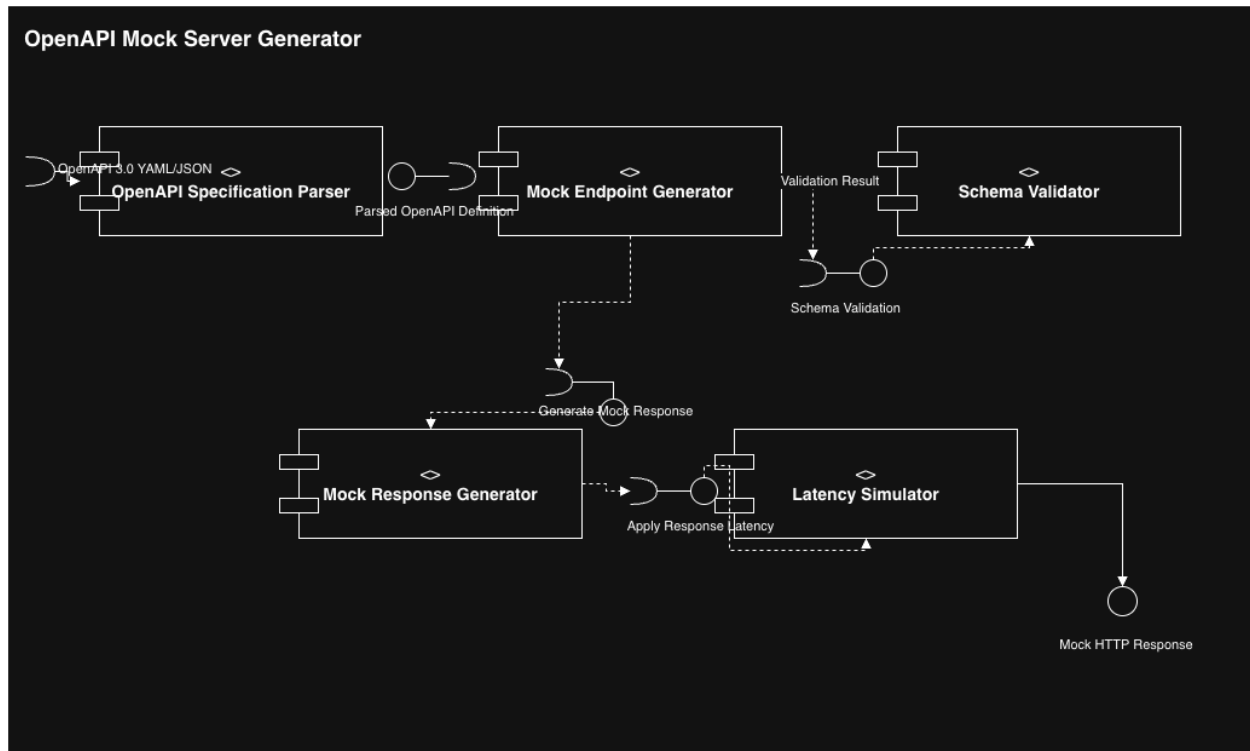
Lab-3 : Component Modelling and Architectural pattern Selection

Problem Statement #41 – Open API Mock Server Generator

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1)UML Component Diagram



2)Architecture Selection – OpenAPI Mock Server Generator

Architecture Selection

We chose **Microservices Architecture** for the OpenAPI Mock Server Generator.

Architectural Choice and Justification

Reason 1 – Independent Scalability:

The system must sustain **1,000 mock API requests per second**. Microservices allow individual components that handle API requests and generate responses to be scaled independently based on their workload.

Reason 2 – Separation of Responsibilities:

The system performs several distinct functions, including OpenAPI specification parsing, dynamic mock endpoint generation, schema validation, mock response generation, and response latency simulation. Separating these functions into independent services provides clear responsibilities and makes the system easier to maintain.

Security Advantage

The **Schema Validator** can reject malformed or invalid API requests before they are processed further. This supports the security requirement that invalid requests should be rejected without exposing sensitive server information.

Performance Benefit

Microservices allow heavily used components of the mock server to be **independently scaled**, helping the system achieve the required throughput of **1,000 requests per second**. The Latency Simulator can also apply configurable response latency to simulate different API response conditions.